

Data Integrity and Validation -- Guided Notes

Unit tp-5.5 | CompTIA Network+ N10-009 Obj. FC0-U71 5.5 | N10-008 FC0-U71 5.5

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Data integrity means data is _____ (correct), _____ (complete), and _____ (not changed without authorization). Organizations achieve data integrity through _____ (enforcing data types and required fields), _____ (detecting unauthorized changes), and access controls.

Question 2

Input validation ensures data entered into a system meets _____ criteria before being processed. Frontend validation in a browser gives _____ feedback but can be bypassed. Server-side validation is _____ because it is enforced in code the user cannot modify. Both should be used together.

Question 3

A _____ constraint in a database prevents NULL (empty) values in a column. A _____ constraint ensures all values in a column are different. A _____ constraint limits which values are permitted (e.g., grade must be A, B, C, D, or F). A _____ constraint enforces the relationship between two tables using keys.

Question 4

A checksum or _____ (hash) function produces a fixed-length value from input data. If even one _____ in the original data changes, the hash changes completely. This property allows detection of _____ during transmission or storage. Common hash functions include _____ and _____.

Question 5

Data _____ is the process of ensuring data is formatted correctly for analysis (fixing errors, removing duplicates, standardizing formats). _____ data (incorrect, inconsistent, or incomplete records) leads to unreliable analysis results. The rule is: _____ in, _____ out.

Question 6

Referential integrity ensures that a foreign key value in one table must match a _____ key that _____ in the referenced table, or be NULL. If a parent record is deleted, the DBMS can _____ delete all child records (CASCADE), or _____ the deletion if child records exist (RESTRICT).

Question 7

Data _____ involves documenting data assets -- what data exists, where it is stored, who owns it, and its _____ level. Strong governance practices include _____ policies that specify how long data must be retained and when it must be _____. Retaining data longer than necessary increases _____ risk.

Question 8 -- Real World Application

REAL WORLD: A student builds a web form that accepts a user's age as a text field. A malicious user enters 'abc' as their age, which crashes the database query. Another user enters -5 as their age, which the system accepts. Identify both problems and describe the server-side validation rules needed to fix them.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-5-5-data-integrity-and-validation/tp-5-5-data-integrity-and-validation-answer-key.pdf
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